

Apache Airflow Hands-On Exercise

Exercise: Daily Student Task Workflow

- By Santhosh Kumar V S

This screenshot shows the Apache Airflow interface with the 'Dags' tab selected. The 'student_daily_tasks_dag' is listed with a status of 'Success'. The latest run was on 2026-05-07 at 14:51:47, and the next run is scheduled for 2026-05-08 at 05:30:00. The DAG is configured with a cron schedule of '0 0 * * *'.

student_daily_tasks_dag
Schedule
Latest Run
Next Run
0 0 * * *
2026-05-07 14:51:47
2026-05-08 05:30:00

This screenshot shows the 'Dag Run' view for the 'student_daily_tasks_dag'. The run is labeled 'scheduled_2026-05-07T00:00:00+00:00' and has a status of 'Success'. The logical date is 2026-05-07 05:30:00, and the run type is 'Scheduled'. The start date is 2026-05-07 14:51:48, and the end date is 2026-05-07 14:51:57. The duration is 00:00:09.443, and the DAG version is v1.

Logical Date	Run Type	Start Date	End Date	Duration	Dag Version(s)
2026-05-07 05:30:00	Scheduled	2026-05-07 14:51:48	2026-05-07 14:51:57	00:00:09.443	v1

Task Instances

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
sleep_task		Success	2026-05-07 14:51:56	1	PythonOperator	00:00:01.330
complete_assignment_task		Success	2026-05-07 14:51:55	1	PythonOperator	00:00:00.287
attend_class_task		Success	2026-05-07 14:51:54	1	PythonOperator	00:00:01.300
wake_up_task		Success	2026-05-07 14:51:48	1	PythonOperator	00:00:09.883

Apache Airflow Hands-On Exercise

Exercise: Daily Student Task Workflow

- By Santhosh Kumar V S

```
root@ubuntu:~$ docker exec -it airflow bash
airflow@bcaf24b852a7:/opt/airflow$ cd /opt/airflow/dags
airflow@bcaf24b852a7:/opt/airflow/dags$ cat > student_daily_tasks_dag.py
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime

def wake_up():
    print("Student woke up at 6 AM")

def attend_class():
    print("Student attended Python class")

def complete_assignment():
    print("Student completed Airflow assignment")

def sleep():
    print("Student went to sleep at 10 PM")

with DAG(
    dag_id="student_daily_tasks_dag",
    start_date=datetime(2024, 1, 1),
    schedule="@daily",
    catchup=False
) as dag:

    wake_up_task = PythonOperator(
        task_id="wake_up_task",
        python_callable=wake_up
    )

    attend_class_task = PythonOperator(
        task_id="attend_class_task",
        python_callable=attend_class
    )
```

```
with DAG(
    dag_id="student_daily_tasks_dag",
    start_date=datetime(2024, 1, 1),
    schedule="@daily",
    catchup=False
) as dag:

    wake_up_task = PythonOperator(
        task_id="wake_up_task",
        python_callable=wake_up
    )

    attend_class_task = PythonOperator(
        task_id="attend_class_task",
        python_callable=attend_class
    )

    complete_assignment_task = PythonOperator(
        task_id="complete_assignment_task",
        python_callable=complete_assignment
    )

    sleep_task = PythonOperator(
        task_id="sleep_task",
        python_callable=sleep
    )

    wake_up_task >> attend_class_task >> complete_assignment_task >> sleep_task
airflow@bcaf24b852a7:/opt/airflow/dags$ exit
exit
root@ubuntu:~$ docker restart airflow
airflow
root@ubuntu:~$ █
```